

Rational Design, Synthesis & Validation of a Novel Ferulic Acid Derivative Targeting COX-2 in Breast Cancer

An end-to-end computational and wet-lab drug-discovery case study

Design → Dock → Molecular Dynamics → Green Synthesis → Characterisation

Lead candidate	FA19 — 4-(2-carboxyvinyl)-2-methoxyphenyl 4-methoxybenzoate (C ₁₈ H ₁₆ O ₆)
Target	Cyclooxygenase-2 (COX-2), PDB 3LN1
Docking affinity (MOE)	-8.3 kcal/mol (vs -5.5 parent control)
MD stability (50 ns)	Ca RMSD 1.47 Å; key H-bonds >80% occupancy
Synthesis	Microwave-assisted, one-pot; up to 93% crude yield
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Executive summary

This case study documents the complete discovery cycle for **FA19**, a novel ferulic acid derivative designed to inhibit cyclooxygenase-2 (COX-2), a validated target in breast-cancer-associated inflammation. The work spans computation and the wet lab: a 24-compound library was screened *in silico*, the lead was docked into COX-2 and shown to occupy the selectivity pocket, its complex was validated over a 50-nanosecond molecular-dynamics simulation, and the molecule was then synthesised by a green, microwave-assisted route and confirmed by spectroscopy. The project demonstrates a pharmacist executing a full Design-Test-Analyse-Make (DMTA) workflow end to end.

Key result. FA19 introduces a bulky 4-methoxybenzoyl group that reaches the COX-2 hydrophobic accessory pocket, improving predicted binding from -5.5 to **-8.3 kcal/mol**. Across a 50 ns simulation the protein backbone is stable (C-alpha RMSD 1.47 +/- 0.17 Å) and the ligand maintains hydrogen bonds to the canonical COX active-site residues **Arg120 (95.2%)**, **Tyr355 (87.6%)** and **Ser530 (80.4%)**. The molecule is highly synthesizable (SA score 1.85, 0 Lipinski violations) and was prepared at up to **93% crude yield** in a one-pot microwave reaction, establishing wet-lab feasibility for scale-up.

Workflow at a glance

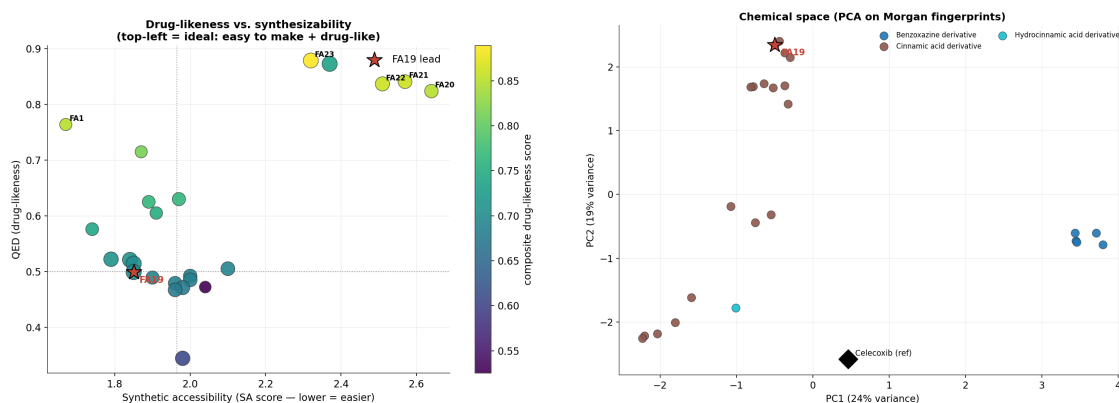
Phase	What was done	Headline outcome
Design	Virtual screen of 24 derivatives; properties, drug-likeness, chemical space	22/24 pass drug-likeness gate; FA19 clean + synthesizable
Test	Dock into COX-2 (MOE; whole-library AutoDock Vina cross-check)	FA19 -8.2 (Vina) vs -8.3 (MOE) kcal/mol; occupies COX-2 selectivity pocket
Analyse	50 ns molecular dynamics of the FA19-COX2 complex (YASARA)	Stable complex; active-site H-bonds >80%; binding triad rigid (RMSF)
Make	Green, microwave-assisted one-pot synthesis; NMR + FTIR	Up to 93% crude; (E)-configuration confirmed by NMR/FTIR

Phase 1 | Design — virtual screening

A focused library of 24 ferulic acid derivatives (alkyl/aralkyl esters, amides, phenolic acylations, bulky aroyl esters, ring-fused analogues) was profiled with RDKit: molecular weight, lipophilicity, polar surface area, hydrogen-bond counts, drug-likeness (QED), synthetic accessibility, and PAINS/Brenk structural alerts. Candidates were pushed through a transparent funnel, and 2048-bit Morgan fingerprints were used for PCA and Butina clustering to map chemical-space diversity.

22 of 24 compounds cleared the drug-likeness gate (≤ 1 Lipinski violation, Veber-compliant, no PAINS alerts). The lead **FA19** ranks **#15 of 24** on drug-likeness alone (QED 0.50, SA score 1.85, 0 Lipinski violations).

FA19 is selected for binding, not for being the most drug-like molecule. This is the correct logic of a screening funnel: filter out developability liabilities first, then let docking choose the lead from a clean pool. Critically, the ferulic series sits far from celecoxib in chemical space, so FA19 is a genuinely novel chemotype rather than a coxib analogue.



Left: drug-likeness vs. synthesizability (FA19 starred). Right: PCA chemical space; FA19 (star) clusters with bulky aroyl esters, far from celecoxib (diamond).

Full screening methodology and the complete ranked table are in the companion [FA19_virtual_screening_report.pdf](#).

Phase 2 | Test — molecular docking

COX-2 selectivity arises from a hydrophobic accessory pocket that COX-1 lacks. FA19 was designed so that its 4-methoxybenzoyl group reaches into this pocket. Docking in MOE (Molecular Operating Environment) lowered the predicted binding energy from approximately **-5.5 kcal/mol** for the parent control to **-8.3 kcal/mol** for FA19 — a substantial gain consistent with the additional pocket contact.

To avoid depending on a single commercial score, the binding was cross-validated with a reproducible, open-source **AutoDock Vina** pipeline (PDB 3LN1; grid box defined from the co-crystallised celecoxib). The protocol embeds a **celecoxib redocking positive control** — recovering the crystallographic pose (RMSD < 2 Å) is the evidence that the docking set-up is trustworthy before any FA19 number is interpreted. Because Vina and MOE use different scoring functions, agreement within roughly 1 kcal/mol is the meaningful criterion, not an identical value. Run across the full 24-compound library, the open-source pipeline placed **FA19 at -8.2 kcal/mol (rank 8 of 24)** — within 0.1 kcal/mol of the MOE -8.3 — with the benzoxazines and bulky benzoyl esters as the strongest binders. Two independent engines converging on the same value is far stronger evidence than either alone. The notebooks (Batch_Docking_Library_Vina.ipynb, FA19_COX2_Vina_redocking.ipynb) run end to end on Google Colab.

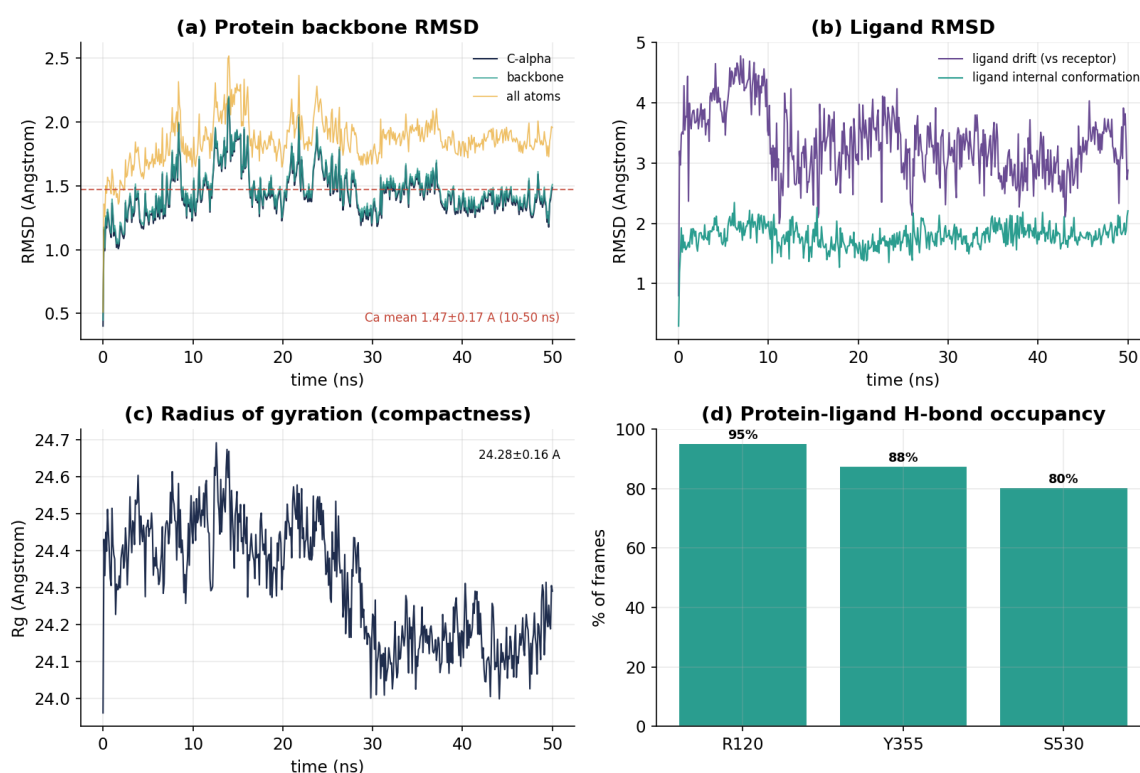
Phase 3 | Analyse — molecular dynamics validation

The FA19-COX2 complex was simulated for **50 ns** in YASARA (501 frames, 100 ps spacing). Docking gives a static snapshot; molecular dynamics tests whether the predicted binding survives thermal motion and solvation. It does.

Metric	Mean +/- SD (10-50 ns)	Interpretation
C-alpha RMSD	1.47 +/- 0.17 A	stable protein fold
Backbone RMSD	1.49 +/- 0.16 A	no large conformational drift
Ligand internal RMSD	1.77 +/- 0.17 A	ligand conformation stable
Ligand drift (vs receptor)	3.17 +/- 0.43 A	settles into a stable pose
Radius of gyration	24.28 +/- 0.16 A	compact, no unfolding
Active-site RMSF	Arg120 1.01 / Tyr355 0.74 / Ser530 0.6 A	binding triad rigid (< 0.93 A mean)

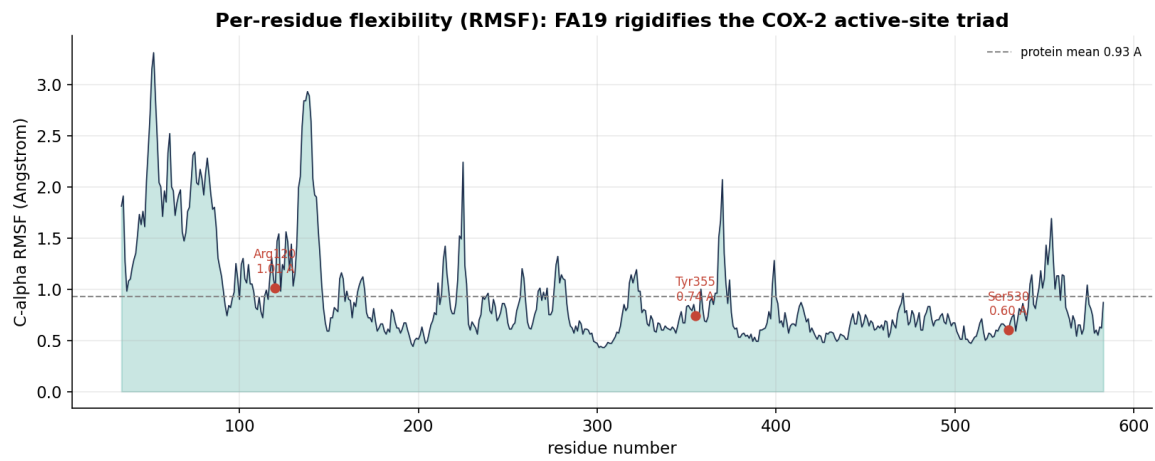
Mechanistically, FA19 binds where a real COX inhibitor should. It maintains hydrogen bonds to **Arg120 (95.2%)** and **Tyr355 (87.6%)** — the constriction-site residues that anchor carboxylate-bearing inhibitors — and to **Ser530 (80.4%)**, the catalytic serine acetylated by aspirin. FA19's own carboxylic acid is the anchor, recapitulating the canonical COX binding mode and holding it for the great majority of the trajectory.

Molecular-dynamics validation of the FA19-COX2 complex (50 ns, YASARA)



Molecular-dynamics summary: (a) protein backbone RMSD, (b) ligand drift vs. internal conformation, (c) radius of gyration, (d) protein-ligand hydrogen-bond occupancy.

Per-residue flexibility (RMSF) closes the loop. The three residues that hydrogen-bond FA19 are also the **least mobile** in the protein — Arg120, Tyr355 and Ser530 fluctuate by only 1.01, 0.74 and 0.6 A respectively, at or below the 0.93 A protein average, while surface loops swing up to ~3 A. FA19 does not merely sit in the pocket; it locks the catalytic triad in place.

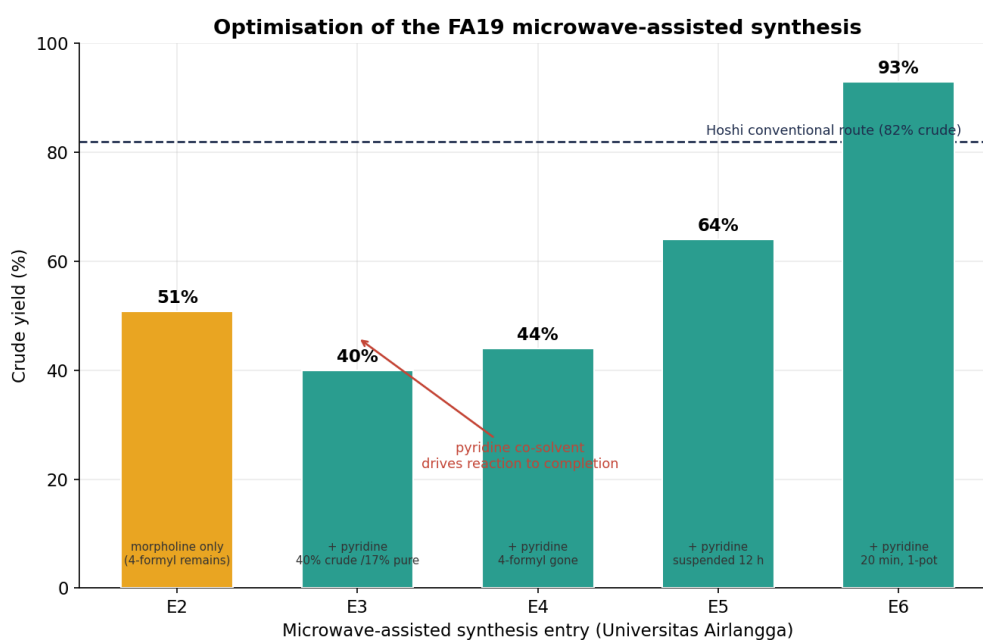


Per-residue C-alpha RMSF over the 50 ns trajectory. Red markers: the active-site triad, all below the protein mean (dashed) — a rigid, well-engaged binding site.

Phase 4 | Make — green synthesis & characterisation

The condensation route was originally established at **Hoshi University (Japan)** using pyrrolidine under conventional conditions (about 82% crude). The chemistry was **transferred to Universitas Airlangga (Indonesia)** and re-developed as a green, **microwave-assisted, one-pot Doebner condensation** of the 4-methoxybenzoyloxy-methoxybenzaldehyde with malonic acid.

The key optimisation insight was that adding **pyridine** as a co-solvent drove the reaction to completion, eliminating residual 4-formyl starting material (tracked by TLC). Tuning time, base equivalents, and a brief aqueous suspension then pushed the crude yield to **93% in a 20-minute reaction** — exceeding the conventional reference while cutting reaction time from hours to minutes.



Crude-yield optimisation across the microwave campaign. E2 (morpholine only, des-methoxy analogue) left starting material unreacted; pyridine addition (E3 onward) drove completion. E3 also gave 17.1% pure material after recrystallisation. Dashed line: Hoshi conventional route.

Structural confirmation

The product structure and (E)-geometry were confirmed by ^1H NMR (399.78 MHz) and FTIR. The diagnostic signals below are taken from the purified material (CDCl_3); coupling constants are calculated directly from the spectrum.

Chemical shift (ppm)	Multiplicity / J	Assignment
8.17	d, J = 8 Hz (2H)	benzoyl ArH ortho to C=O
7.76	d, J = 16 Hz (1H)	-CH=CH-Ar (trans)
7.17-7.19	m	methoxyphenyl ring H
6.99	d, J = 8 Hz (2H)	benzoyl ArH ortho to OCH ₃
6.43	d, J = 16 Hz (1H)	=CH-COOH (trans)
3.91 / 3.86	s, s (3H each)	two OCH ₃ groups

The two vinyl doublets with J = 16 Hz are decisive: they confirm the trans (E) configuration of the cinnamoyl double bond. The two para-coupled aromatic doublets (J = 8 Hz) confirm the 1,4-disubstituted 4-methoxybenzoyl ring, and the two methoxy singlets confirm both methoxy groups. (The CDCl_3 sample also showed minor aliphatic impurity peaks, consistent with the lab notes on purification; the DMSO-d_6 spectrum gave a matching aromatic/vinyl pattern.)

FTIR (purified material) corroborates the structure: an ester carbonyl near **1724 cm⁻¹**, a conjugated carboxylic-acid carbonyl near **1685 cm⁻¹** with a broad O-H envelope (2500-3300 cm⁻¹), conjugated and aromatic C=C around **1602 and 1514 cm⁻¹**, ester C-O-C stretches at **1160-1260 cm⁻¹**, and a trans-alkene out-of-plane band near **980 cm⁻¹** — independently consistent with the (E)-configured aryl-ester product.

From bench to pilot — feasibility & scale-up case

Taken together, the data support advancing FA19 toward a pilot-scale process and an industry conversation. The argument rests on three pillars:

- **Developability.** FA19 is drug-like by construction: molecular weight 328, cLogP 3.0, zero Lipinski violations, no PAINS/Brenk alerts, and a favourable synthetic-accessibility score (1.85).
- **Target engagement.** Strong predicted binding (-8.3 kcal/mol) is backed by a 50 ns molecular-dynamics simulation showing a stable complex and persistent hydrogen bonds to the canonical COX-2 active-site residues — evidence the binding is mechanistically meaningful, not a docking artefact.
- **Manufacturability.** The synthesis is green and scalable: a one-pot, microwave-assisted condensation reaching 93% crude yield in 20 minutes, with a simple aqueous work-up (filtration/washing) and recrystallisation for purification. Reaction times in minutes (vs. hours by reflux) translate into throughput and energy savings, and the route has already survived a successful technology transfer between two institutions.

Next development steps would be in vitro COX-2 inhibition and selectivity assays (COX-2 vs COX-1), cytotoxicity in breast-cancer cell lines, purity/stability characterisation for the pilot batch, and a process-optimisation study to improve isolated (not just crude) yield through purification.

Honest limitations

- **No experimental bioactivity yet.** Binding is predicted (docking) and dynamically stable (MD), but no measured IC50/inhibition or cell data are included. These are the necessary next experiments.
- **QED reporting.** The RDKit default QED for FA19 is ~0.50; a different tool/weighting can give other values. Any reported figure should state its method.
- **Confirm the redock control each run.** The whole-library Vina docking is done (FA19 -8.2 vs MOE -8.3); the celecoxib redock positive control (RMSD < 2 Å) for its FA19 number to be trusted.
- **Yield is crude.** The headline 93% is crude yield; isolated/pure yield after recrystallisation is lower (e.g. 17.1% in one entry) and is a clear process-optimisation target.
- **Screening library shown here is representative.** The exact professor-supplied FA1-FA24 conclusions follow once those SMILES are loaded into the pipeline.

Methods & reproducibility

Design: RDKit descriptors, QED, Ertl-Schuffenhauer SA score, Lipinski/Veber/Egan filters, PAINS+Brenk catalogues; Morgan fingerprints (radius 2, 2048 bits) with PCA and Butina clustering (scikit-learn). **Test:** MOE docking; reproducible AutoDock Vina + Meeko pipeline on PDB 3LN1 with a celecoxib positive control. **Analyse:** 50 ns YASARA molecular dynamics; RMSD/Rg/H-bond analysis in Python. **Make:** microwave-assisted one-pot Doebner condensation; structure by 1H NMR (399.78 MHz) and FTIR. All computational code, data, figures, and the docking notebook are provided in the project repository so every number reproduces.

Project repository: virtual-screening pipeline (screen.py, plots.py), MD analysis (md_analysis.py), synthesis figure (synthesis_optimization.py), QSAR scaffold (qsar_template.py), docking notebook (notebooks/), data/ and figures/. Generated with RDKit, scikit-learn, AutoDock Vina, YASARA, MOE.